

CHAPTER 11 INFINITE SEQUENCES AND SERIES

11.1 Sequences

- A sequence

$$a_1, a_2, a_3, a_4, \dots$$

is a list of numbers written in a definite order. We denote it by $\{a_n\}_{n=1}^{\infty}$ or simply $\{a_n\}$. a_n is the n th term.

- A sequence may be specified by giving enough initial terms to establish a pattern, such as

$$1, 4, 7, 10, 13, \dots \quad \text{or} \quad \{1, 4, 7, 10, 13, \dots\},$$

or by an explicit formula for the n th term, such as

$$a_n = 3n - 2, \quad n \geq 1 \quad \text{or} \quad \{3n - 2\}_{n=1}^{\infty},$$

or by a recursive formula

$$a_1 = 1, \quad a_n = a_{n-1} + 3, \quad n \geq 2.$$

- **Example** Write down the first 4 terms of the following sequences:

(a) $a_n = 1 - \frac{1}{n}, \quad n \geq 1$ $\{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\}$

(b) $b_n = 1 + (-1)^n \frac{1}{n} \quad n \geq 1$ $\{0, \frac{3}{2}, \frac{2}{3}, \frac{5}{4}, \dots\}$

(c) $\left\{(-1)^n + \frac{1}{n}\right\}_{n=3}^{\infty}$ $\{-\frac{2}{3}, \frac{5}{4}, -\frac{4}{5}, \frac{7}{6}, \dots\}$
 $a_3 \quad a_4 \quad a_5 \quad a_6$

(d) $\{0.999\}_{n=1}^{\infty}$

$\{0.999, 0.999, 0.999, 0.999, \dots\}$

- **Example** Find a formula for the general term of the sequence

$$\left\{ \frac{3}{5}, -\frac{4}{25}, \frac{5}{125}, -\frac{6}{625}, \frac{7}{3125}, \dots \right\}$$

$$a_n = (-1)^{n-1} \frac{n+2}{5^n}, \quad n \geq 1$$

- **Example** The following is the famous Fibonacci sequence

$$\{1, 1, 2, 3, 5, 8, 13, 21, \dots\}$$

Express the sequence using a recursive formula.

$$a_1 = 1$$

$$a_2 = 1$$

$$a_{n+2} = a_n + a_{n+1}, \quad n \geq 1$$

- **Definition**

A sequence $\{a_n\}$ is said to have a **limit** L or **converge** to L , denoted by

$$\lim_{n \rightarrow \infty} a_n = L,$$

if the values of a_n gets arbitrary close to L when n is large enough.

Otherwise, we say $\{a_n\}$ **diverges**.

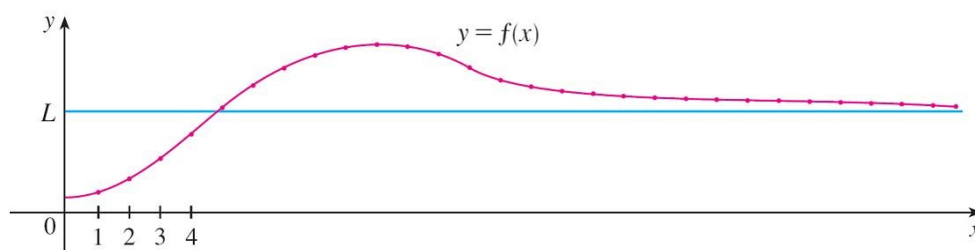


- **Theorem**

If $\lim_{x \rightarrow \infty} f(x) = L$, then $\lim_{n \rightarrow \infty} f(n) = L$.

$\uparrow$
real numbers

$\uparrow$
integers



- **Example** Find $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}}$.

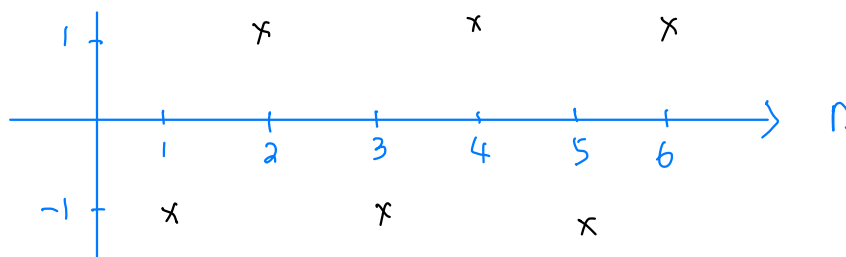
sequence $\left\{ \frac{1}{\sqrt{n}} \right\}_{n=1}^{\infty}$

$$\lim_{\substack{\textcircled{n} \rightarrow \infty \\ \text{integers}}} \frac{1}{\sqrt{n}} = \lim_{\substack{\textcircled{x} \rightarrow \infty \\ \text{real numbers}}} \frac{1}{\sqrt{x}} = 0$$

The sequence $\left\{ \frac{1}{\sqrt{n}} \right\}_{n=1}^{\infty}$ converges to 0.

- **Example** Does the sequence $\{(-1)^n\}$ converge? If yes, to what number?

$\{-1, 1, -1, 1, -1, 1, \dots\}$



The sequence alternates between -1 and 1 , clearly it does not converge.

• Properties of Limits of Sequences

Let $\{a_n\}$ and $\{b_n\}$ be convergent sequences and c a constant. Then

1. $\lim_{n \rightarrow \infty} c = c$
2. $\lim_{n \rightarrow \infty} ca_n = c \lim_{n \rightarrow \infty} a_n$
3. $\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$
4. $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$
5. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$, if $\lim_{n \rightarrow \infty} b_n \neq 0$.

- **Example** Find $\lim_{n \rightarrow \infty} \frac{3n^2}{7n^2 + 1}$.

Sequence $\left\{ \frac{3n^2}{7n^2 + 1} \right\}_{n=1}^{\infty}$, converge?

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{3n^2}{7n^2 + 1} & \times \frac{\frac{1}{n^2}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{3}{7 + \frac{1}{n^2}} \\
 & = \frac{3}{7 + 0} \\
 & = \frac{3}{7}
 \end{aligned}$$

The sequence $\left\{ \frac{3n^2}{7n^2 + 1} \right\}_{n=1}^{\infty}$ converges to $\frac{3}{7}$.

- **Example** Does the sequence $\left\{ \frac{n}{\sqrt{en+2}} \right\}$ converge? If yes, to what number? ∞
 $n=1$

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt{en+2}} \quad \times \quad \frac{\frac{1}{\sqrt{n}}}{\frac{1}{\sqrt{n}}}$$

$$= \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{e + \frac{2}{n}}}$$

$$= \infty \quad \text{The sequence } \left\{ \frac{n}{\sqrt{en+2}} \right\} \text{ diverges.}$$

- **Example** Does the sequence $\left\{ \frac{\ln n}{n} \right\}$ converge? If yes, to what number?

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

$$\stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1}$$

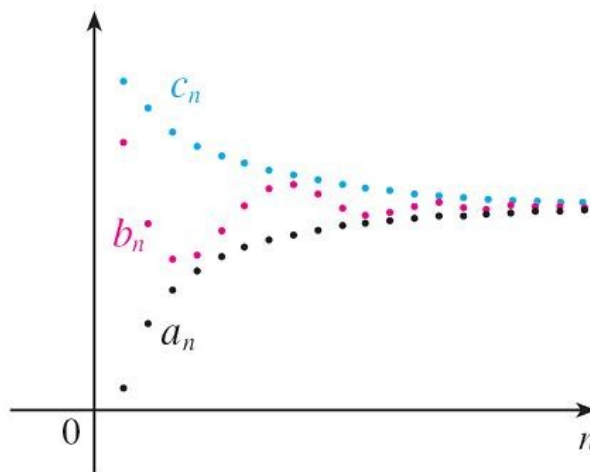
$$= \lim_{x \rightarrow \infty} \frac{1}{x}$$

$$= 0$$

$$\text{The sequence } \left\{ \frac{\ln n}{n} \right\} \text{ converges to } 0.$$

- Squeeze Theorem**

Suppose that $\{a_n\}$ and $\{c_n\}$ both converge to L , and that $a_n \leq b_n \leq c_n$ for all $n \geq N$, then $\lim_{n \rightarrow \infty} b_n = L$.



- Example** Find $\lim_{n \rightarrow \infty} \frac{\sin^3 n}{n}$.

Sequence $\left\{ \frac{\sin^3 n}{n} \right\}_{n=1}^{\infty}$ converge?

$$-1 \leq \sin n \leq 1$$

$$-1 \leq \sin^3 n \leq 1$$

$$-\frac{1}{n} \leq \frac{\sin^3 n}{n} \leq \frac{1}{n}$$

The sequence

$\left\{ \frac{\sin^3 n}{n} \right\}$ converges

to 0.

$$\lim_{n \rightarrow \infty} \left(-\frac{1}{n} \right) = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

By Squeeze Theorem, $\lim_{n \rightarrow \infty} \frac{\sin^3 n}{n} = 0$

- **Example** Show that if $\lim_{n \rightarrow \infty} |a_n| = 0$, then $\lim_{n \rightarrow \infty} a_n = 0$.

useful !

$$-|a_n| \leq a_n \leq |a_n|$$

$$\lim_{n \rightarrow \infty} -|a_n|$$

$$\lim_{n \rightarrow \infty} |a_n| = 0$$

$$= - \lim_{n \rightarrow \infty} |a_n|$$

By Squeeze Theorem,

$$= -0$$

$$\lim_{n \rightarrow \infty} a_n = 0.$$

$$= 0$$

- **Example** Find $\lim_{n \rightarrow \infty} \frac{(-1)^n}{\sqrt{n}}$.

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^n}{\sqrt{n}} \right| = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}}$$

$$= 0$$

$$\therefore \lim_{n \rightarrow \infty} \frac{(-1)^n}{\sqrt{n}} = 0.$$

- **Example** Find $\lim_{n \rightarrow \infty} \frac{n!}{n^n}$.

$$0 \leq \frac{1 \times 2 \times 3 \times \cdots \times n}{n \times n \times n \times \cdots \times n} \leq \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} 0 = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

By Squeeze Theorem, $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$

- **Theorem**

If $\lim_{n \rightarrow \infty} a_n = L$ and the function f is continuous at L , then

$$\lim_{n \rightarrow \infty} f(a_n) = f(L)$$

- **Example** Find $\lim_{n \rightarrow \infty} \sin\left(\frac{\pi}{n}\right)$.

$f(x) = \sin x$
is continuous

$$\lim_{n \rightarrow \infty} \sin\left(\frac{\pi}{n}\right)$$

$$= \sin\left(\lim_{n \rightarrow \infty} \frac{\pi}{n}\right)$$

$$= \sin\left(\pi \lim_{n \rightarrow \infty} \frac{1}{n}\right)$$

$$= \sin 0$$

$$= 0$$

- **Definition**

- A sequence $\{a_n\}$ is called **increasing** if $a_n < a_{n+1}$ for all n .
- A sequence $\{a_n\}$ is called **nondecreasing** if $a_n \leq a_{n+1}$ for all n .
- A sequence $\{a_n\}$ is called **decreasing** if $a_n > a_{n+1}$ for all n .
- A sequence $\{a_n\}$ is called **nonincreasing** if $a_n \geq a_{n+1}$ for all n .
- A sequence that is increasing or decreasing is called a **strictly monotonic sequence**.
- A sequence that is nondecreasing or nonincreasing is called a **monotonic sequence**.

- **Definition**

- A sequence $\{a_n\}$ is **bounded above** if there is a number M such that

$$a_n \leq M \quad \forall n \geq 1$$

- A sequence $\{a_n\}$ is **bounded below** if there is a number m such that

$$a_n \geq m \quad \forall n \geq 1$$

- If a sequence is bounded above and bounded below, we say that it is a **bounded sequence**.

- **Example** Show that the sequence $\left\{\frac{n}{n+2}\right\}$ is increasing. Is the sequence bounded?

$$\begin{aligned}
 a_{n+1} - a_n &= \frac{n+1}{n+3} - \frac{n}{n+2} \\
 &= \frac{(n+1)(n+2) - n(n+3)}{(n+2)(n+3)} \\
 &= \frac{\cancel{n^2} + \cancel{3n} + 2 - \cancel{n^2} - \cancel{3n}}{(n+2)(n+3)}
 \end{aligned}$$

$$= \frac{2}{(n+2)(n+3)} > 0 \quad \text{for } n \geq 1$$

$\therefore \left\{\frac{n}{n+2}\right\}$ is increasing.

$$a_1 = \frac{1}{3}$$

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{n}{n+2} &\times \frac{\frac{1}{n}}{\frac{1}{n}} \\
 &= \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{2}{n}}
 \end{aligned}$$

$$= 1$$

$$\frac{1}{3} \leq a_n < 1$$

- **Example** Show that the sequence $\left\{\frac{n}{n^2+1}\right\}$ is decreasing. Is the sequence bounded?

Can show $a_{n+1} - a_n < 0$.

$$\text{Let } f(x) = \frac{x}{x^2+1}$$

real numbers

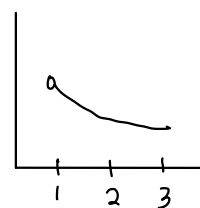
$$f'(x) = \frac{(1)(x^2+1) - (2x)(x)}{(x^2+1)^2}$$

$$= \frac{1 - x^2}{(x^2+1)^2} < 0, \quad x > 1$$

$\therefore \left\{\frac{n}{n^2+1}\right\}$ is decreasing.

$$a_1 = \frac{1}{2}$$

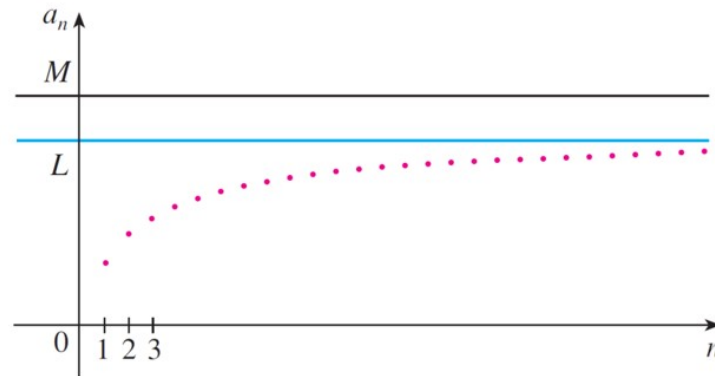
$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{n}{n^2+1} &\times \frac{\frac{1}{n^2}}{\frac{1}{n^2}} \\
 &= \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{1 + \frac{1}{n^2}} = \frac{0}{1+0} = 0
 \end{aligned}$$



$$0 < a_n \leq \frac{1}{2}$$

• Monotonic Sequence Theorem

Every bounded, monotonic sequence is convergent.



- Example** Suppose you know that $\{a_n\}$ is an monotonic increasing sequence and all its terms lie between the numbers $\frac{1}{3}$ and 1. Explain why the sequence has a limit. What can you say about the value of the limit?

$$\frac{1}{3} \leq a_n \leq 1 \quad \text{bounded}$$

The sequence is given to be monotonic and bounded, by the Monotonic Sequence Theorem, the sequence is convergent. By the definition of convergent, the sequence has a limit.

$$\frac{1}{3} < \lim_{n \rightarrow \infty} a_n \leq 1$$

$$\frac{1}{3} \leq a_n < 1$$

11.2 Series

- An **infinite series** is of the form

$$a_1 + a_2 + a_3 + a_4 + \cdots$$

which is also denoted by $\sum_{n=1}^{\infty} a_n$ or $\sum a_n$.

- Example** $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = ?$

Guess :

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = 1$$

n	Sum of first n terms
1	0.50000000
2	0.75000000
3	0.87500000
4	0.93750000
5	0.96875000
6	0.98437500
7	0.99218750
10	0.99902344
15	0.99996948
20	0.99999905
25	0.99999997

$S_1 = \frac{1}{2}$
 $S_2 = \frac{1}{2} + \frac{1}{4}$
 $S_3 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$
 S_4
 S_5
 S_{25}

- The n^{th} partial sum of an infinite series $\sum a_n$, denoted by S_n , is given by

$$S_n = a_1 + a_2 + \cdots + a_n = \sum_{i=1}^n a_i$$

- Example** Show that the n^{th} partial sum S_n of a geometric series $a + ar + ar^2 + ar^3 + \cdots$ is $\frac{a(1-r^n)}{1-r}$.

$$\begin{aligned}
 S_n &= a + ar + ar^2 + \cdots + ar^{n-1} \\
 -) \quad r S_n &= ar + ar^2 + \cdots + ar^{n-1} + ar^n \\
 \hline
 (1-r) S_n &= a - ar^n \\
 \Rightarrow S_n &= \frac{a(1-r^n)}{1-r}
 \end{aligned}$$

- **Definition**

Given a series $\sum_{n=1}^{\infty} a_n$, let S_n denote its n^{th} partial sum:

$$S_n = \sum_{i=1}^n a_i = a_1 + a_2 + \cdots + a_n$$

If the sequence $\{S_n\}$ is convergent, and $\lim_{n \rightarrow \infty} S_n = S$ exists as a real number, then the series $\sum a_n$ is called **convergent** and we write

$$\sum_{n=1}^{\infty} a_n = \lim_{n \rightarrow \infty} S_n = S$$

The number S is called the **sum** of the series.

If the sequence $\{S_n\}$ diverges, then the series diverges. A divergent series has no sum.

- **Example** Show that a geometric series $a + ar + ar^2 + ar^3 + \cdots$ converges if and only if $-1 < r < 1$.

n^{th} partial sum, $S_n = \frac{a(1-r^n)}{1-r}$

$$\begin{aligned}
 S &= \lim_{n \rightarrow \infty} S_n \\
 &= \lim_{n \rightarrow \infty} \frac{a(1-r^n)}{1-r} \\
 &= \frac{a(1 - \lim_{n \rightarrow \infty} r^n)}{1-r}
 \end{aligned}$$

$\lim_{n \rightarrow \infty} r^n = \begin{cases} 0 & , -1 < r < 1 \\ 1 & , r = 1 \\ \text{d.n.e.} & , r > 1 \text{ or } r < -1 \end{cases}$

S converge iff $-1 < r < 1$

$$= \frac{a(1-0)}{1-r} = \frac{a}{1-r}, \quad -1 < r < 1$$

Consider $r = 1$,

$$S = a + a(1) + a(1)^2 + a(1)^3 + \dots$$

$$= a + a + a + a + \dots$$

$$= \infty$$

- **Example** Find the sum of the series

$$5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \dots$$

Geometric series : $a = 5$, $r = \frac{-\frac{10}{3}}{5} = -\frac{2}{3}$

↑

$$5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \dots$$

Check :

$$= \frac{5}{1 - (-\frac{2}{3})}$$

$$\text{color: red; } -1 < r < 1$$

$$= 3$$

- **Example** Is the series $\sum_{n=1}^{\infty} 2^{2n} 3^{1-n}$ convergent or divergent?

$$\sum_{n=1}^{\infty} 2^{2n} 3^{1-n} = \sum_{n=1}^{\infty} 3 \cdot \frac{4^n}{3^n}$$

$$= \sum_{n=1}^{\infty} 3 \cdot \left(\frac{4}{3}\right)^n$$

Geometric series : $a = 3\left(\frac{4}{3}\right) = 4$, $r = \frac{4}{3}$

Since $|r| \geq 1$, the series is divergent.

- **Example** A drug is administered to a patient at the same time every day. Suppose that the concentration of the drug is C_n (measured in mg/mL) after the injection on the n^{th} day. Before the injection the next day, only 30% of the drug remains in the bloodstream and the daily doses raises the concentration by 0.2 mg/mL.

(a) Find the concentration after 3 days.

$$\text{First : } C_1 = 0.2 \text{ mg/mL}$$

$$\text{Second : } C_2 = 0.2 + 0.3(0.2) = 0.26 \text{ mg/mL}$$

$$\text{Third : } C_3 = 0.2 + 0.3(0.26) = 0.278 \text{ mg/mL}$$

(b) What is the concentration after the n^{th} dose?

$$\begin{aligned}
 C_n &= 0.2 + 0.3 C_{n-1} \\
 &= 0.2 + 0.3 (0.2 + 0.3 C_{n-2}) \\
 &= 0.2 + 0.3 (0.2) + (0.3)^2 C_{n-2} \\
 &= 0.2 + (0.3)(0.2) + (0.3)^2(0.2) + \dots + (0.3)^{n-1} C_1 \\
 &= 0.2 + (0.3)(0.2) + (0.3)^2(0.2) + \dots + (0.3)^{n-1}(0.2) \\
 &= \frac{0.2(1-0.3^n)}{1-0.3} = \frac{2}{7} (1-0.3^n) \quad \text{geometric series: } a = 0.2 \\
 &\quad \quad \quad r = 0.3
 \end{aligned}$$

(c) What is the limiting concentration?

$$\begin{aligned}
 \lim_{n \rightarrow \infty} C_n &= \lim_{n \rightarrow \infty} \frac{2}{7} (1-0.3^n) \\
 &= \frac{2}{7} (1 - \lim_{n \rightarrow \infty} 0.3^n) = \frac{2}{7} (1-0) = \frac{2}{7} \text{ mg/mL}
 \end{aligned}$$